

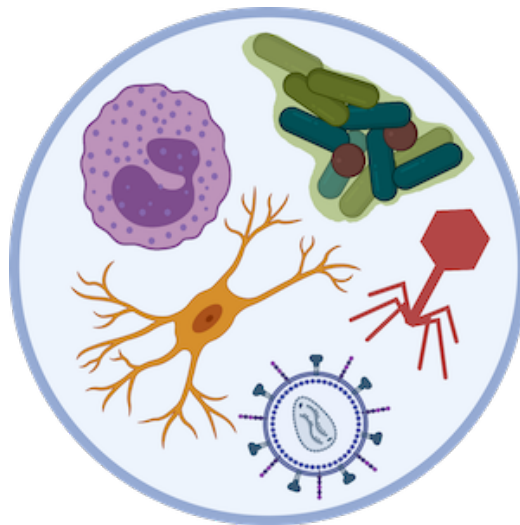
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IGCSE Double Award Science Biology Revised June 2026

Unit 1 Cells & Microbes Student Booklet



Lesson 1: Characteristics of Living Things

Lesson 2: Animal and Plant Cells

Lesson 3: Diffusion

Lesson 4 & 5: Osmosis

Lesson 6: Active Transport

Lesson 7: Pathogens

Lesson 8: Revision Resources

Double Award Biology - Cells and Microbes Syllabus Statements

1 The nature and variety of living organisms

(a) Characteristics of living organisms

Students should:

1.1 understand how living organisms share the following characteristics:

- they require nutrition
- they respire
- they excrete their waste
- they respond to their surroundings
- they move
- they control their internal conditions
- they reproduce
- they grow and develop.

(b) Variety of living organisms

Students should:

1.2 describe the common features shown by eukaryotic organisms: plants, animals, fungi and protocists

Plants: these are multicellular organisms; their cells contain chloroplasts and are able to carry out photosynthesis; their cells have cellulose cell walls; they store carbohydrates as starch or sucrose. Examples include flowering plants, such as a cereal (for example, maize), and a herbaceous legume (for example, peas or beans).

Animals: these are multicellular organisms; their cells do not contain chloroplasts and are not able to carry out photosynthesis; they have no cell walls; they usually have nervous co-ordination and are able to move from one place to another; they often store carbohydrate as glycogen. Examples include mammals (for example, humans) and insects (for example, housefly and mosquito).

Fungi: these are organisms that are not able to carry out photosynthesis; their body is usually organised into a mycelium made from thread-like structures called hyphae, which contain many nuclei; some examples are single-celled; their cells have walls made of chitin; they feed by extracellular secretion of digestive enzymes onto food material and absorption of the organic products; this is known as saprotrophic nutrition; they may store carbohydrate as glycogen. Examples include *Mucor*, which has the typical fungal hyphal structure, and yeast, which is single-celled.

Protocists: these are microscopic single-celled organisms. Some, like *Amoeba*, that live in pond water, have features like an animal cell, while others, like *Chlorella*, have chloroplasts and are more like plants. A pathogenic example is *Plasmodium*, responsible for causing malaria.

Students should:

1.3 describe the common features shown by prokaryotic organisms such as bacteria

Bacteria: these are microscopic single-celled organisms; they have a cell wall, cell membrane, cytoplasm and plasmids; they lack a nucleus but contain a circular chromosome of DNA; some bacteria can carry out photosynthesis but most feed off other living or dead organisms. Examples include *Lactobacillus bulgaricus*, a rod-shaped bacterium used in the production of yoghurt from milk, and

Pneumococcus, a spherical bacterium that acts as the pathogen causing pneumonia.

Double Award Biology - Cells and Microbes Syllabus Statements

- 1.4 understand the term pathogen and know that pathogens may include fungi, bacteria, protoctists or viruses
- Viruses: these are not living organisms. They are small particles, smaller than bacteria; they are parasitic and can reproduce only inside living cells; they infect every type of living organism. They have a wide variety of shapes and sizes; they have no cellular structure but have a protein coat and contain one type of nucleic acid, either DNA or RNA. Examples include the tobacco mosaic virus that causes discolouring of the leaves of tobacco plants by preventing the formation of chloroplasts, the influenza virus that causes 'flu' and the HIV virus that causes AIDS.

2 Structure and functions in living organisms

(a) Level of organisation

Students should:

- 2.1 describe the levels of organisation in organisms: organelles, cells, tissues, organs and systems

(b) Cell structure

Students should:

- 2.2 describe cell structures, including the nucleus, cytoplasm, cell membrane, cell wall, mitochondria, chloroplasts, ribosomes and vacuole
- 2.3 describe the functions of the nucleus, cytoplasm, cell membrane, cell wall, mitochondria, chloroplasts, ribosomes and vacuole
- 2.4 know the similarities and differences in the structure of plant and animal cells

(d) Movement of substances into and out of cells

Students should:

- 2.15 understand the processes of diffusion, osmosis and active transport by which substances move into and out of cells
- 2.16 understand how factors affect the rate of movement of substances into and out of cells, including the effects of surface area to volume ratio, distance, temperature and concentration gradient
- 2.17 *practical: investigate diffusion and osmosis using living and non-living systems*

(h) Transport

Students should:

- 2.51 understand why simple, unicellular organisms can rely on diffusion for movement of substances in and out of the cell

Cells and Microbes – Glossary of Key Words

Keyword	Definition
Active Transport	The movement of substances from a more dilute solution to a more concentrated solution (against a concentration gradient). This requires the use of energy from respiration.
ATP	ATP (adenosine triphosphate) is the energy-carrying molecule used in cells because it can release energy very quickly. Energy is released from ATP when the end phosphate is removed.
Bacteria	Microscopic single-celled organisms. They may produce toxins that damage tissues and make us feel ill. Examples include <i>Salmonella</i> , which causes food poisoning, and the pathogen that causes gonorrhea.
Cell	The basic building block of all living organisms.
Cell (Plasma) Membrane	A partially permeable membrane that surrounds the cytoplasm and controls the movement of substances into and out of the cell.
Cell Wall	A rigid structure found in plant cells and bacterial cells that strengthens the cell and provides support.
Cellulose	The main structural component in the cell walls of plant cells.
Chlorophyll	A green pigment found in chloroplasts which absorbs light for photosynthesis.
Chloroplast	An organelle that contains the pigment chlorophyll which absorbs light so the plant can carry out photosynthesis.
Concentrated Solution	A solution that contains a large amount of solute.
Concentration	A measure of the amount of solute in a solution.
Concentration Gradient	The difference in concentration between two solutions on either side of a membrane.
Cytoplasm	The jelly-like substance in which sub-cellular structures are found and where most chemical reactions occur.
Diffusion	The net movement of particles from an area of higher concentration to an area of lower concentration (down the concentration gradient).
Dilute Solution	A solution that contains a small amount of solute.
Electron Microscope	A microscope that uses a beam of electrons to produce a magnified image of an object. This type of microscope has high magnification and resolution.
Eukaryotic Cell	A type of cell that has a nucleus.
Flagellum	A tail-like structure that enables bacteria and sperm cells to move.
Fungi	A large group of eukaryotic organisms that contain single-celled yeasts, moulds and mushrooms.

Light Microscope	A microscope that uses lenses and visible light to produce a magnified image of an object. This type of microscope has low magnification and resolution.
Magnification	The amount that an image of something is scaled up when viewed through a microscope. Magnification = image size ÷ actual size
Mitochondria	Structures in the cytoplasm of all cells where aerobic respiration takes place (singular is mitochondrion).
Nucleus	An organelle that contains the genetic material of the cell and controls the activities of the cell.
Organ	A group of tissues that work together to perform a specific function.
Organ System	A group of organs working together to perform a function.
Organelle	A structure found within a cell. Also called a sub-cellular structure.
Organism	A single living thing. Can be multicellular or single-celled.
Osmosis	The diffusion of water from a dilute solution to a concentrated solution through a partially permeable membrane.
Pathogen	A microorganism that causes infectious disease.
Prokaryotic Cell	A simple cell that does not have a nucleus – the DNA is free in the cytoplasm.
Protoctist	A eukaryotic, usually single-celled organism of the kingdom Protista.
Ribosome	A sub-cellular structure that carries out protein synthesis using the genetic code from DNA.
Selectively Permeable Membrane	A membrane that allows small molecules (e.g. water and certain solutes) to pass across it but does not allow the passage of large molecules.
Surface Area To Volume Ratio	The internal volume of the object compared with the amount of area where it contacts its environment.
Tissue	A group of cells with a similar structure and function working together.
Vacuole	An organelle filled with sap to keep the cell rigid enough to support the plant.
Virus	An ultramicroscopic infectious non-cellular organism that can replicate inside the cells of living hosts, with negative consequences.

Lesson 1: Characteristics of Living Organisms

What makes something alive? State what is meant by the acronym MRS C GREN.

M

R

S

C

G

R

E

N

True or False?

1. All living organisms have eyes.
2. All living organisms breathe.
3. All living organisms need nutrition.
4. All living organisms can reproduce.
5. All living organisms do digestion.

Exam Practice

Q1. All living organisms have certain characteristics.

(a) Describe what is meant by the following characteristics:

(i) respiration

(2)

.....

.....

.....

(ii) control of their internal conditions

(2)

.....

.....

.....



There are over 2000 types of jellyfish around the world. Jellyfish swim by rhythmic pulsations of the umbrella. The tentacles on a jellyfish are covered with millions of stinging cells. Jellyfish usually feed on small fish that get caught in their tentacles or sometimes even other jellyfish.

Jellyfish take oxygen into their cells directly from the water around them. Jellyfish don't actually have brains but they do have something called a nerve net which can coordinate responses. Most Jellyfish release sperm or eggs directly into the water around them. Jellyfish have no blood, bones or hearts. But some of them can do incredible things like glow in the dark.

Jellyfish are some of the oldest living creatures on Earth and have probably been around on the planet for over 600 million years. Jellyfish numbers are increasing as they are one of the few living organisms that are thriving with warmer more acidic oceans.

Q2. Suggest and explain 3 pieces of evidence from the passage that show that jellyfish are a type of living organism.

(3)

.....

.....

.....

.....

Q3. Suggest how jellyfish carry out excretion.

(1)

.....

.....

Q4: Some jellyfish can clone themselves to make identical copies. Suggest how this also shows they are living organisms.

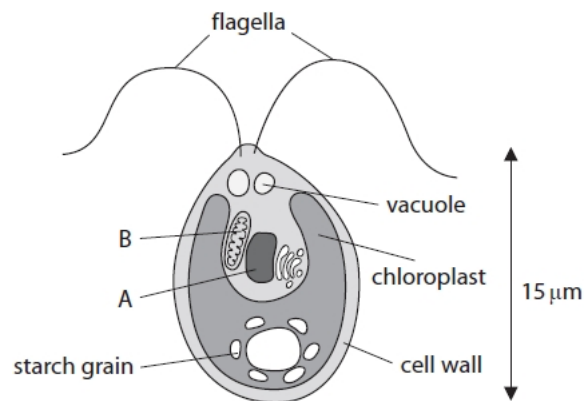
(1)

.....

.....

Kingdom	Characteristics	Examples	Cell Diagram
Animal			
Plant			
Fungi			
Bacteria			
Protocist			

Q1. The diagram shows *Chlamydomonas*, a single-celled protist that is found in water.



(a) Name the structures labelled A and B.

(2)

A

B

(b) (i) *Chlamydomonas* has some structures that are normally found in plant cells, but not found in animal cells.

Name two of these structures.

(2)

1

2

(ii) State why *Chlamydomonas* is not classified as a plant.

(1)

.....

.....

(c) Explain how *Chlamydomonas* obtains energy.

(2)

.....

.....

.....

.....

(d) *Chlamydomonas* has flagella.

Suggest a function of these flagella.

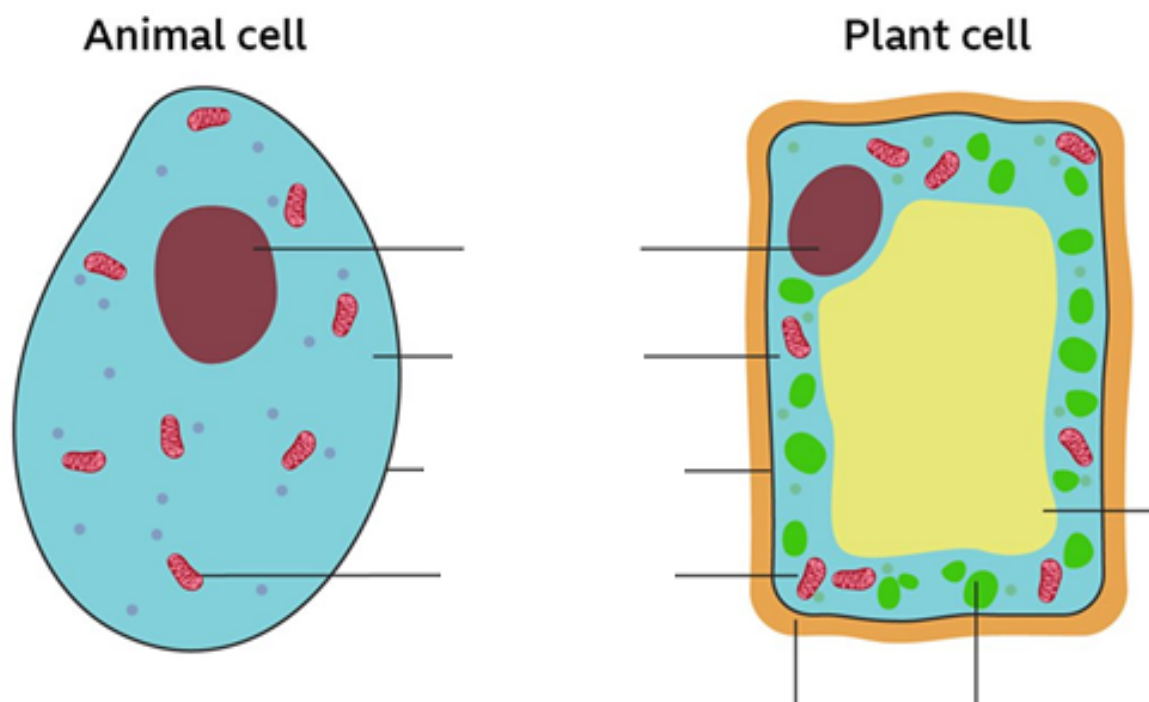
(1)

.....

.....

Lesson 2: Animal and Plant Cells

Label the cell structures:



Match the cell structure to its function:

Cell Structure
Cell Wall
Vacuole
Cell Membrane
Chloroplast
Cytoplasm
Nucleus
Mitochondria

Function
Stores water and nutrients
Controls what comes in and out of the cell
Holds the DNA
Where photosynthesis happens
Provides strength and support for the cell
Where energy is released from food
Where chemical reactions happen

What does organelle mean?

.....

Which organelles are found in both animal and plant cells?

.....

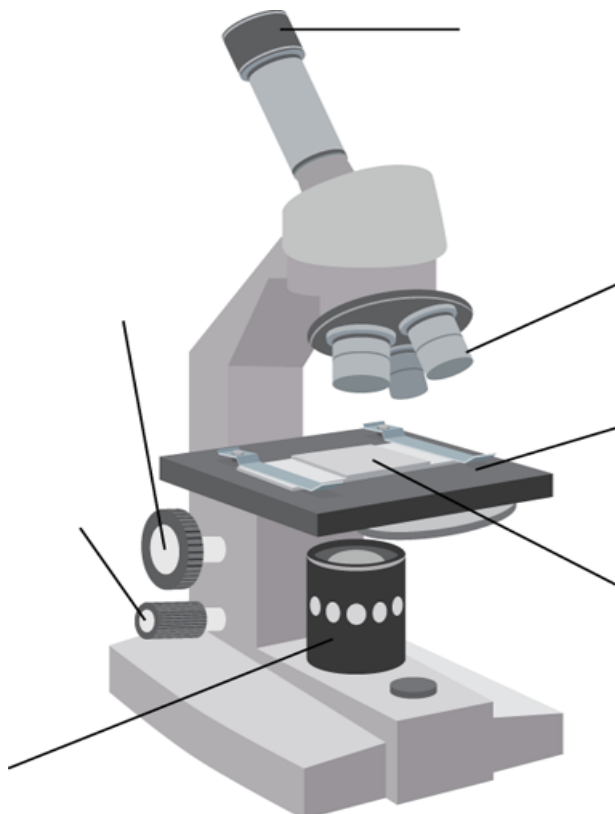
Which organelles can be found in only plant cells?

.....

Parts of a Light Microscope

Label the microscope using the following words:

coarse focus, fine focus, stage, light, slide, objective lens, eyepiece



Explain the functions of the following parts of the microscope:

Lamp:

Focusing knob:

Stage:

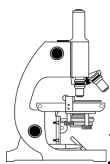
Objective lens:

Which objective lens would you use to find your specimen?

.....

Which objective lens would you use to observe detail?

.....



Using a Microscope

(Never touch the surfaces of the eyepiece or objective lens, use specialized lens tissue if they are dirty)

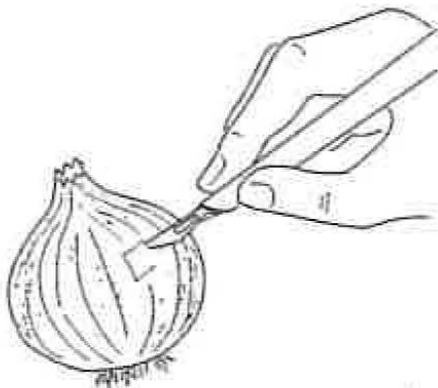
Number the following statements in the correct order:

Order	Instructions
	Once your specimen is in focus, you can increase the magnification by turning the nosepiece so that a higher power objective lens is above the slide
	Use the rough focusing knob to move the stage to its lowest position
	Place your slide on to the stage
	Look through the eyepiece lens
	Locate your specimen using the rough focusing knob (you may need to adjust the position of the slide by moving the stage)
	Adjust the focus for the enlarged image by using the fine focusing knob – do NOT use the rough focusing knob at this point as you could damage the lens or crack the slide!
	Turn the nosepiece so that the low power objective lens clicks into position above the hole in the stage
	Secure your slide with the stage clips
	Switch on your microscope

Once you have the steps in the correct order, you are ready to begin looking at real slides!

Looking at plant cells

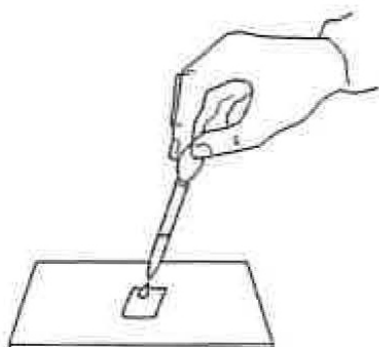
Before you look at onion cells under the microscope, you must peel off a very thin layer.

Step 1

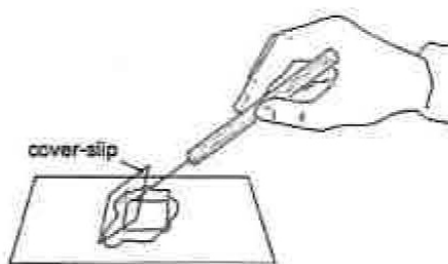
- **Carefully** cut out a small piece of onion tissue.

Step 2

- Use forceps or your finger nails to peel off the inner surface (this looks like tissue paper).

Step 3

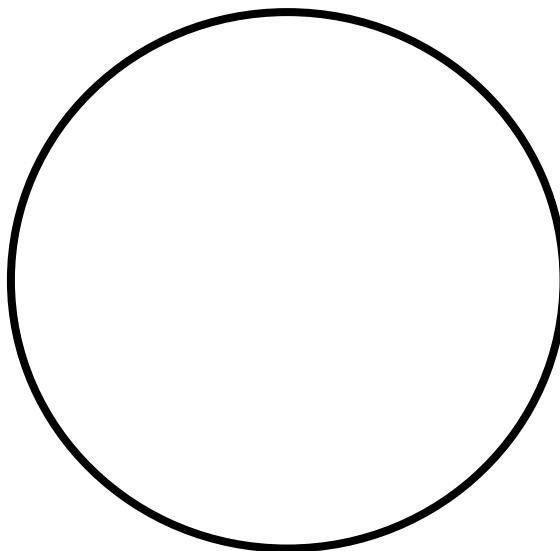
- Put the piece of onion 'skin' flat on a slide and add two drops of iodine solution.

Step 4

- Gently lower the cover-slip onto the slide using a mounted needle.
- Now look at your onion cells under the microscope.

Draw what your onion cells look like here:

Don't forget to label any organelles.



Looking at animal cells

To avoid any chance of infection carry out these instructions carefully.



Step 1



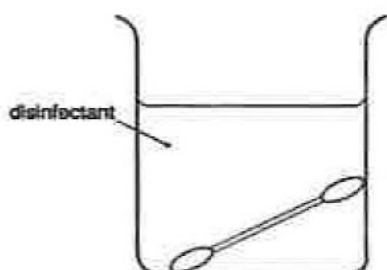
- Your teacher will give you a cotton bud from a freshly opened pack.
- Gently wipe the lining of your cheek with one end of the cotton bud.

Step 2



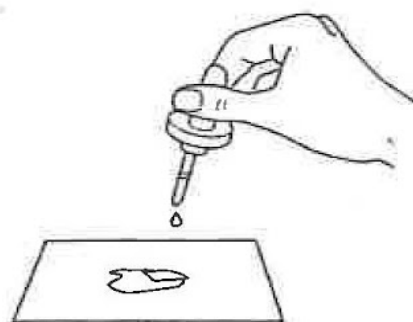
- Smear the cotton bud over the centre of a slide.

Step 3



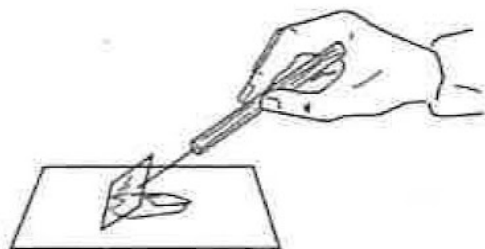
- Immediately dispose of the cotton bud into a beaker of disinfectant provided by your teacher.

Step 4



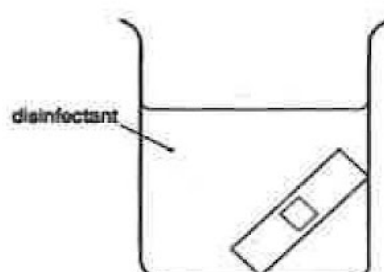
- Put a drop of methylene blue stain on top of the smear.

Step 5



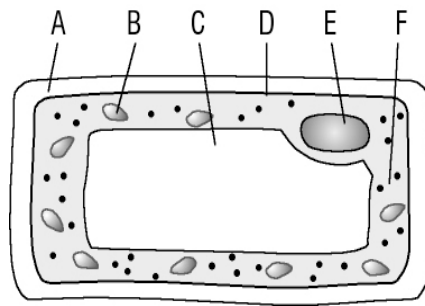
- Place a cover-slip on top and look for cells under the microscope at high power.
- Draw some of your cheek cells (2 or 3) and label your drawing.

Step 6



- At the end of the activity put the slide with its cover-slip into a beaker of disinfectant.

Q1 The diagram is of a cell from the leaf of a plant.



(a) Name the structures **D**, **E** and **F**.

.....

(3)

(b) (i) What is the name of structure **A**?

.....(1)

(ii) What material is structure **A** made of?

.....(1)

(c) (i) What is the name of structure **C**?

.....(1)

(ii) What is the liquid it contains called?

.....(1)

(d) Structure **B** is a chloroplast. What is its function?

.....

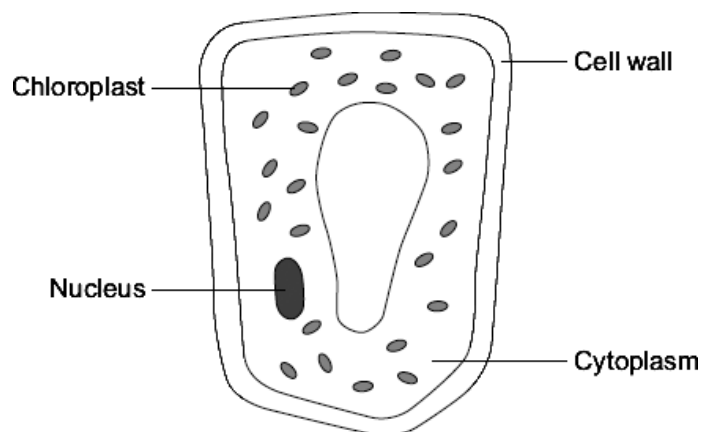
(2)

(e) Name two different structures that are found within the material labelled **F**.

.....

(2)

Q2. The diagram shows a plant cell from a leaf.



(a) **List A** gives the names of three parts of the cell.

List B gives the functions of parts of the cell.

Draw a line from each part of the cell in **List A** to its function in **List B**.

List A
Parts of the cell

Nucleus

Cytoplasm

Chloroplast

List B
Functions

Where most of the chemical reactions take place

Absorbs light energy to make food

Strengthens the cell

Controls the activities of the cell

(3)

(b) Respiration takes place in the cell.

Draw a ring around the correct answer to complete the sentence.

All cells use respiration to release

energy

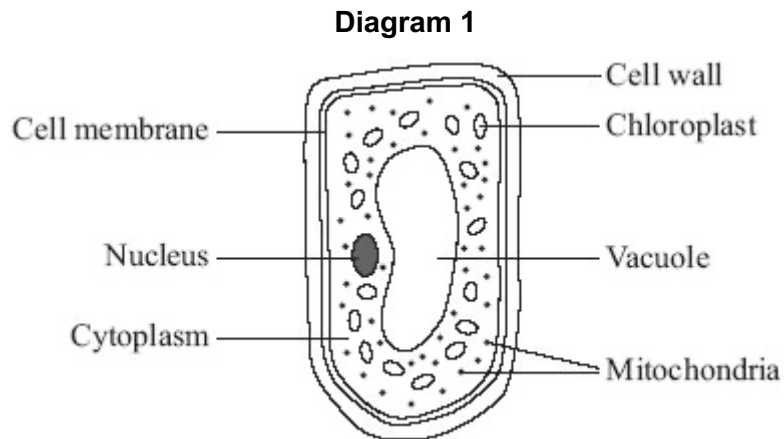
oxygen.

sugar.

(1)

(Total 4 marks)

Q3. Diagram 1 shows a cell from a leaf.



- (a) How is the leaf cell specialised to carry out photosynthesis?

Tick (✓) **one** box.

It has a permanent vacuole.

☐

It has many chloroplasts.

☐

It has cytoplasm.

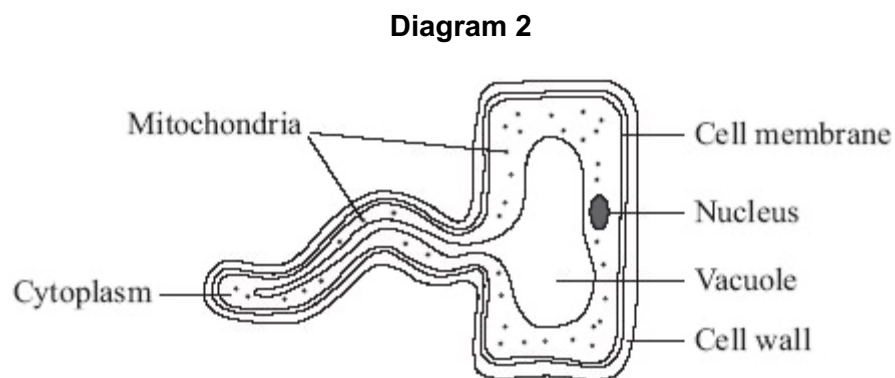
☐

It has many mitochondria.

☐

(1)

- (b) **Diagram 2** shows another type of plant cell.



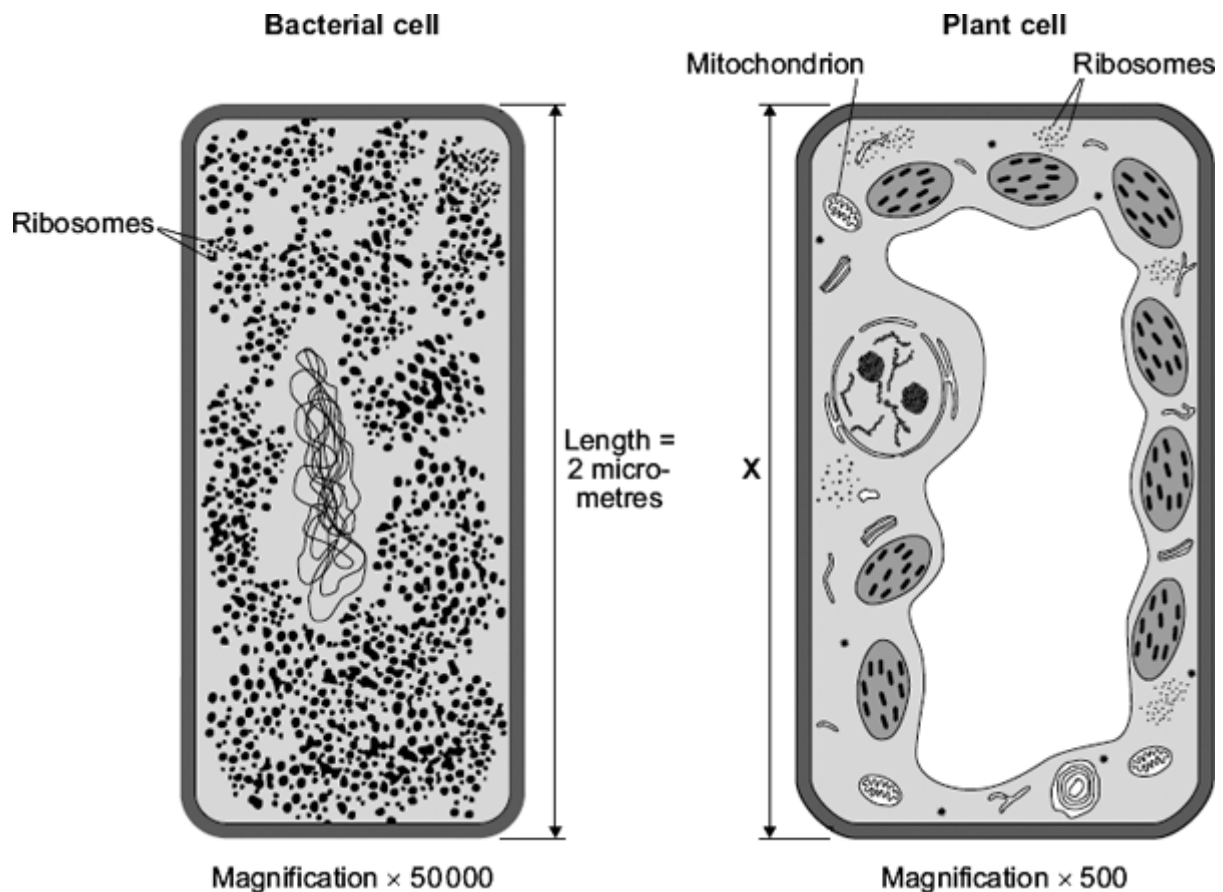
Give **two** ways in which this cell is different from an animal cell.

1. _____

2. _____

(2)

Q4. The diagram shows two cells, a bacterial cell and a plant cell.



- (a) (i) Both the bacterial cell and the plant cell contain ribosomes.

What is the function of a ribosome?

(1)

- (ii) The plant cell contains mitochondria but the bacterial cell does **not** contain mitochondria.

Give **one** other way in which the plant cell is different from the bacterial cell.

(1)

(Total 2 marks)

Lesson 3: Diffusion

What substances does a cell need?

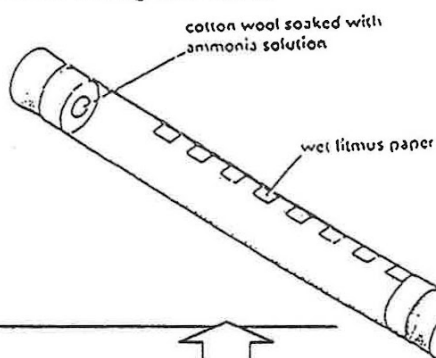
1. What useful substances need to move into a cell?

2. What waste products does a cell need to get rid of?

What is the definition of diffusion?

Proving diffusion!

This apparatus will prove diffusion happens. The wet litmus paper will turn blue as the ammonia solution turns into gas and diffuses along the tube.



1. Q. Can you draw an arrow to show the Direction of diffusion.

Q, Can you label each end of the tube with high concentration and low concentration.

2. So what if we made the concentration of the ammonia solution stronger (more particles!). What do you think will happen to the diffusion speed. (CIRCLE the answer you think in red.)

Speed up

Slow down

Same

3. Now record the time for all the litmus strips to turn blue for the weak ammonia and strong ammonia.

Weak (time) _____ Strong (time) _____

4. Well! Were you right? If NOT circle the correct choice in blue.

Now...can you explain why below?

Concentration affects diffusion speed because.....

Effect of size on uptake by diffusion

SAFETY:

Take care with the solutions used: wear eye protection and rinse splashes off the skin.

PROCEDURE:

- Collect agar cubes of different sizes – 2 cm x 2 cm x 2 cm and 1 cm x 1 cm x 1 cm
- Place the cubes in a beaker and cover with the acid (just enough to cover the surface).
- Start the stop clock.
- Leave the cubes and measure the time taken for the cubes to lose their pink colour completely.
- While you are waiting, complete the table with calculations for each cube. (See student sheet.)

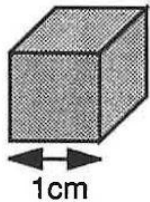
A Length of side of agar cube (cm)	B = A ² Area of one side of cube (cm ²)	C = 6B Total surface area of cube (cm ²)	D = A ³ Volume of cube (cm ³)	E = C÷D Surface area to volume ratio	F Distance travelled by acid (clear) in mm
1					
2					
3					

How to calculate surface area and volume:

Cube A

SA of face = 1cm x 1cm = 1cm²
 SA of cube = 1cm² x 6 = 6cm²
 V of cube = 1cm x 1cm x 1cm = 1cm³
SA/V = 6:1

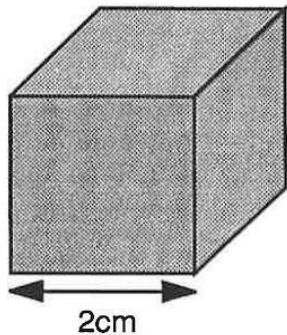
A



Cube B

SA of face = 2cm x 2cm = 4cm²
 SA of cube = 4cm² x 6 = 24cm²
 V of cube = 2cm x 2cm x 2cm = 8cm³
SA/V = 3:1

B



QUESTIONS:

- 1 Compare the surface area of the 1 cm block to the 2 cm block. In each case the length of the side of the block doubles. What happens to the surface area?
.....
- 2 Compare the volume of the 1 cm block to the 2 cm block. In each case the length of the side of the block doubles. What happens to the volume?
.....
- 3 Compare the surface area to volume ratio of the 1 cm block to the 2 cm block. In each case the length of the side of the block doubles. What happens to the surface area to volume ratio?
.....
- 4 Is there any difference in the rate of diffusion as the size of the agar cubes increases?
.....
- 5 The volume of a living organism is proportional to the number of cells in its body. Each cell needs oxygen and nutrients and needs to get rid of metabolic waste products such as carbon dioxide. The smallest organisms can absorb nutrients and get rid of waste by diffusion through their outer membranes. Do you think larger organisms can do the same? Why?
.....
.....
- 6 Should this procedure have a control? If so, what would it be?
.....
.....
- 7 Complete the following paragraph, using the words provided below. Each word may be used once, more than once, or not at all.

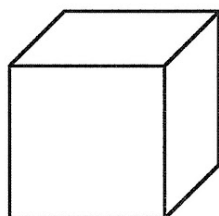
Third	decrease	body surfaces	second	area	diffusion
relative	ratio	volume	surface area	slow	size

Living organisms exchange materials across _____ by _____. Increasing size means that _____ grows faster than _____. Exchanging materials by _____ becomes a very _____ process and not enough material will reach cells buried deep within the body of large organisms. Complex structures like the root system of a plant or the lungs of a mammal provide a large _____ for exchange and short distances for _____.

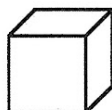
- 8 A student investigated the effect of size on the movement of molecules.

He cut three different sized cubes from a block of clear agar jelly.

Cube A was $2 \times 2 \times 2$ cm. Cube B was $1 \times 1 \times 1$ cm and cube C was $0.5 \times 0.5 \times 0.5$ cm.



cube A



cube B



cube C

The student wore safety glasses and placed the cubes in a beaker of red dye solution for five minutes.

He then poured the solution into another beaker, washed the surface of the cubes and dried them with blotting paper.

He then cut each cube in half and examined the newly cut surfaces.

Diagram 1 shows what the cubes looked like.

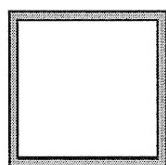


Diagram 1

- a. Why did he wear safety glasses? (1)

.....

.....

- b. Explain how the red dye molecules entered the jelly. (2)

.....

.....

- c. Use a ruler to measure the distance the red dye has entered each cube in diagram 1. (1)

Answer:mm

- d. Calculate the surface area of cube A. (2)

Surface Area=unit

- (ii) Calculate the volume of cube A.

volume= unit

- e. The student was told by his teacher that the cubes have different surface area to volume ratios.

Complete the table by putting one tick (✓) in each row to show whether the statement applies to cube A, B or C. (3)

	Cube A	Cube B	Cube C
Largest surface area			
Largest surface area to volume ratio			
Greatest proportion of cube coloured red			

- f. Explain how this experiment can be used to understand the need to transport systems in larger organisms such as humans. (3)

.....

.....

.....

.....

(Total for Question 8=14 marks)

Lesson 4 Osmosis

Define these terms:

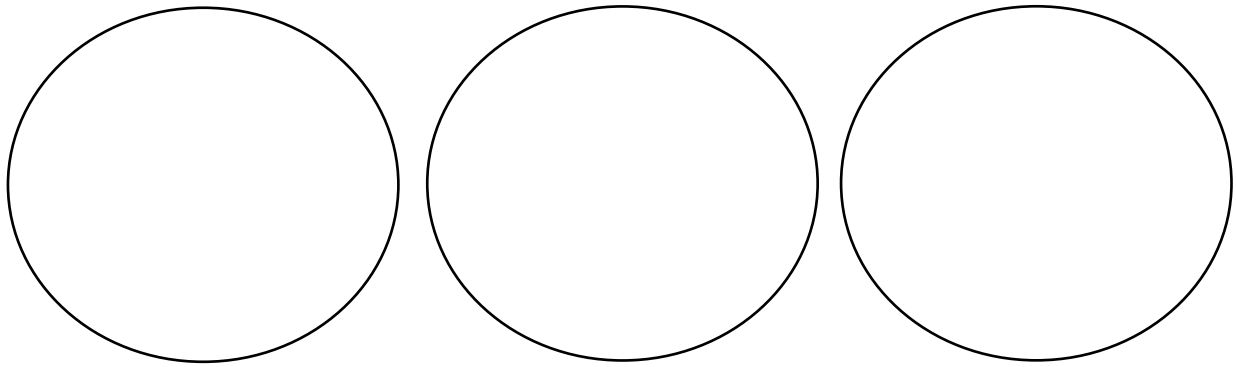
Dilute

.....

Concentrated

.....

Draw a dilute solution, a concentrated solution and a pure water diagram in each of the spaces below.



Define osmosis

.....

.....

.....

What is a partially permeable membrane?

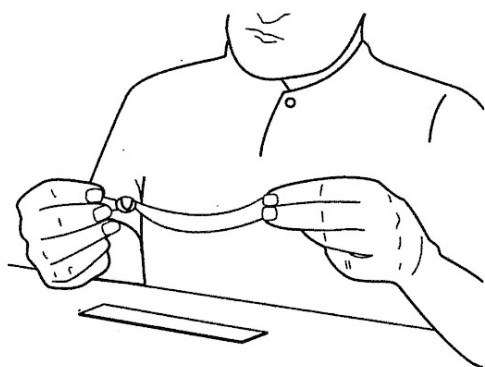
.....

.....

A model cell

Copy and complete this table:

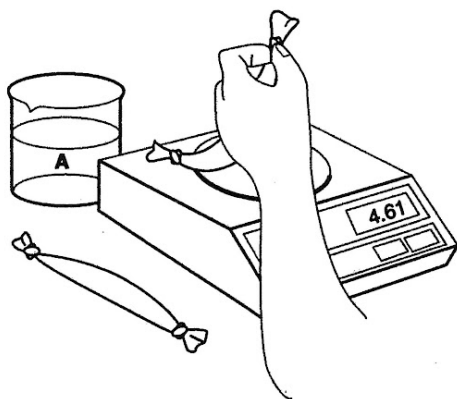
model cell	liquid inside Visking tubing	liquid in beaker	weight of model cell (g)		change in weight of cell (g)
			at start	after 30 mins	
A					
B					

Step 1

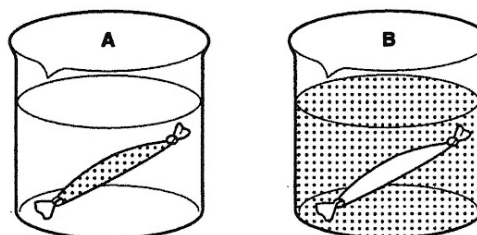
- Cut 2 pieces of Visking tubing 12 cm long. Tie the end of each with cotton.

Step 2

- Fill cell A with sugar solution and tie the open end with cotton.
- Fill cell B with water and tie the open end with cotton.

Step 3

- Wipe the outside of each model cell carefully with a paper towel.
- Weigh each cell and put the reading in the table.

Step 4

- Put cell A into a beaker of water and cell B into a beaker of sugar solution.
- Re-weigh your model cells after 30 minutes.

Questions

1 What did cell A look like after 30 minutes?

2 What has happened to the weight of cell A after 30 minutes?

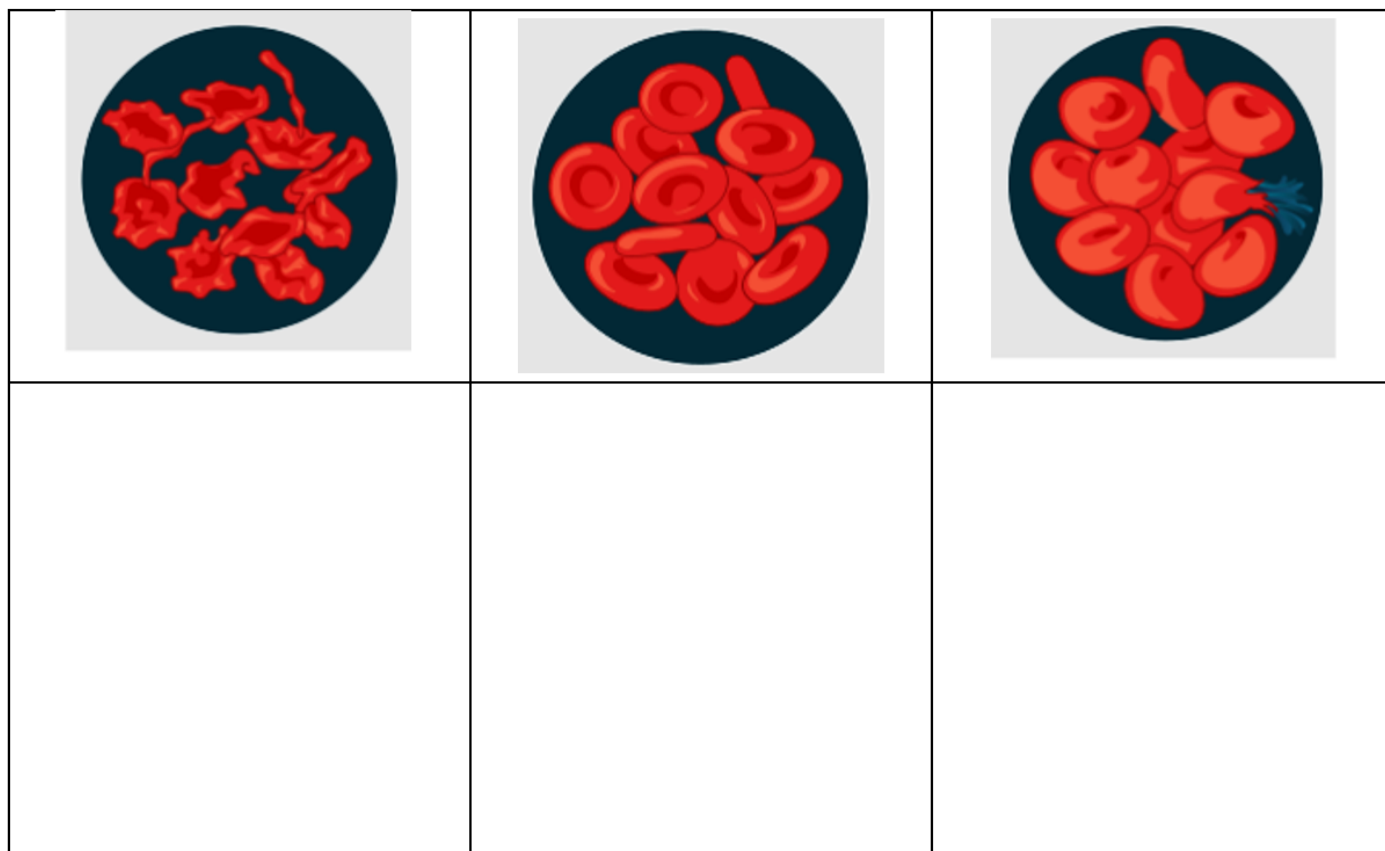
Explain what has happened.

3 What did cell B look like after 30 minutes?

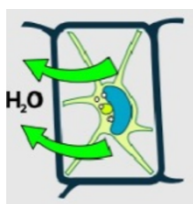
4 What has happened to the weight of cell B after 30 minutes?

Explain what has happened.

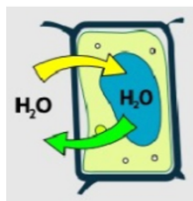
Fill in the cartoon strip to show describe what happens when you place an animal cell into an isotonic (same concentration), a concentrated and a dilute solution.



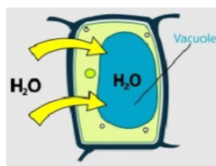
Match the diagram to the correct description of how a plant cell behaves in each type of solution:



Plant cell placed in a concentrated solution. This leads to the movement of water out of the cell. This can lead to plasmolysis.



Plant cell placed in a dilute solution. This leads to the movement of water into the cell causing turgor pressure.

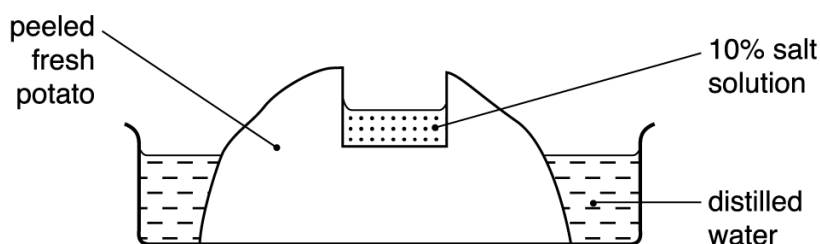


Plant cell placed in a isotonic solution. Water enters and leaves the plant cell at an equal rate, this leaves the plant cell flaccid.

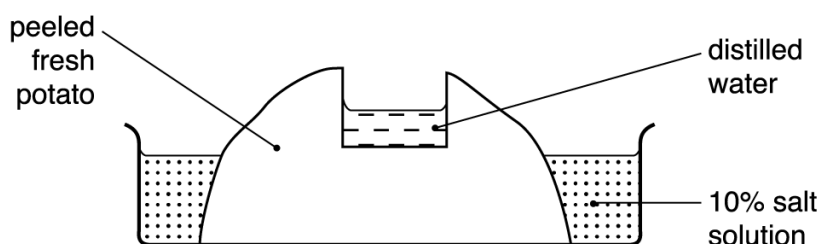
Potato Osmometer

The apparatus in the diagram is used to demonstrate the movement of water through plant tissue.

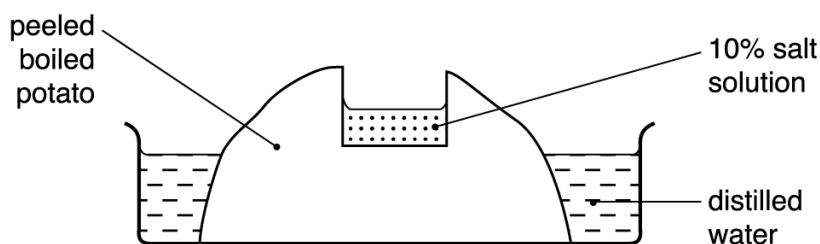
apparatus A



apparatus B



apparatus C



Describe what changes you would expect to observe after leaving the apparatus set up for one hour, and, in each case, explain why the change has occurred.

Apparatus A

Change

Explanation

Apparatus B

Change.

Explanation

Apparatus C

Change

Explanation

Lesson 5: Investigate the effect of sugar solution on plant tissue

In this practical you should:

- Identify independent, dependent and control variables
- Write a risk assessment
- Produce a graph of your results
- Write a conclusion using your results and your scientific knowledge

Equipment

- | | |
|--|----------------------|
| • Potatoes | • Measuring cylinder |
| • Scalpel | • Sticky labels |
| • Ruler | • Test tube rack |
| • 5 boiling tubes | • Kitchen roll |
| • Sugar solution – 0 mol dm ⁻³ , 0.2 mol dm ⁻³ , 0.4 mol dm ⁻³ , 0.6 mol dm ⁻³ , 0.8 mol dm ⁻³ , 1 mol dm ⁻³ | • Balance |

Method

1. Cut the potato into six equally sized cubes (using the ruler) and pat the cubes dry.
2. Use the balance to find the start mass of the cubes.
3. Put the boiling tubes into the rack and put a cube of potato in each tube.
4. Use the measuring cylinder to measure 20cm³ of the first solution into one of the tubes, label the tube to show which solution is in it.
5. Repeat with the other solutions.
6. Leave for the amount of time suggested by your teacher.
7. Remove the cubes from the boiling tubes and re-weigh them.
8. Calculate the change in mass for each cube.

Risk assessment

1. What is the main risk in this experiment?

.....

2. How could you reduce this risk?

.....

Variables

3. What is the independent variable in this experiment?

.....

4. What is the dependent variable in this experiment?

.....

5. What is a control variable in this experiment?

.....

6. How is this variable kept the same?

.....

.....

Results

Concentration (mol dm ⁻³)	Mass of potato at start (g)	Mass of potato at end (g)	Change in mass (g)
0.0			
0.2			
0.4			
0.6			
0.8			
1.0			

Draw a graph of your results on the next page

Analysis

1. Which variable goes on your x-axis?

.....

2. Which variable goes on your y-axis?

.....

3. Why do you calculate % change?

.....

4. In the diagram shown, what is the arrow showing moving out of the potato?

.....

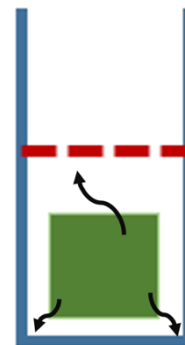
5. What is this process known as?

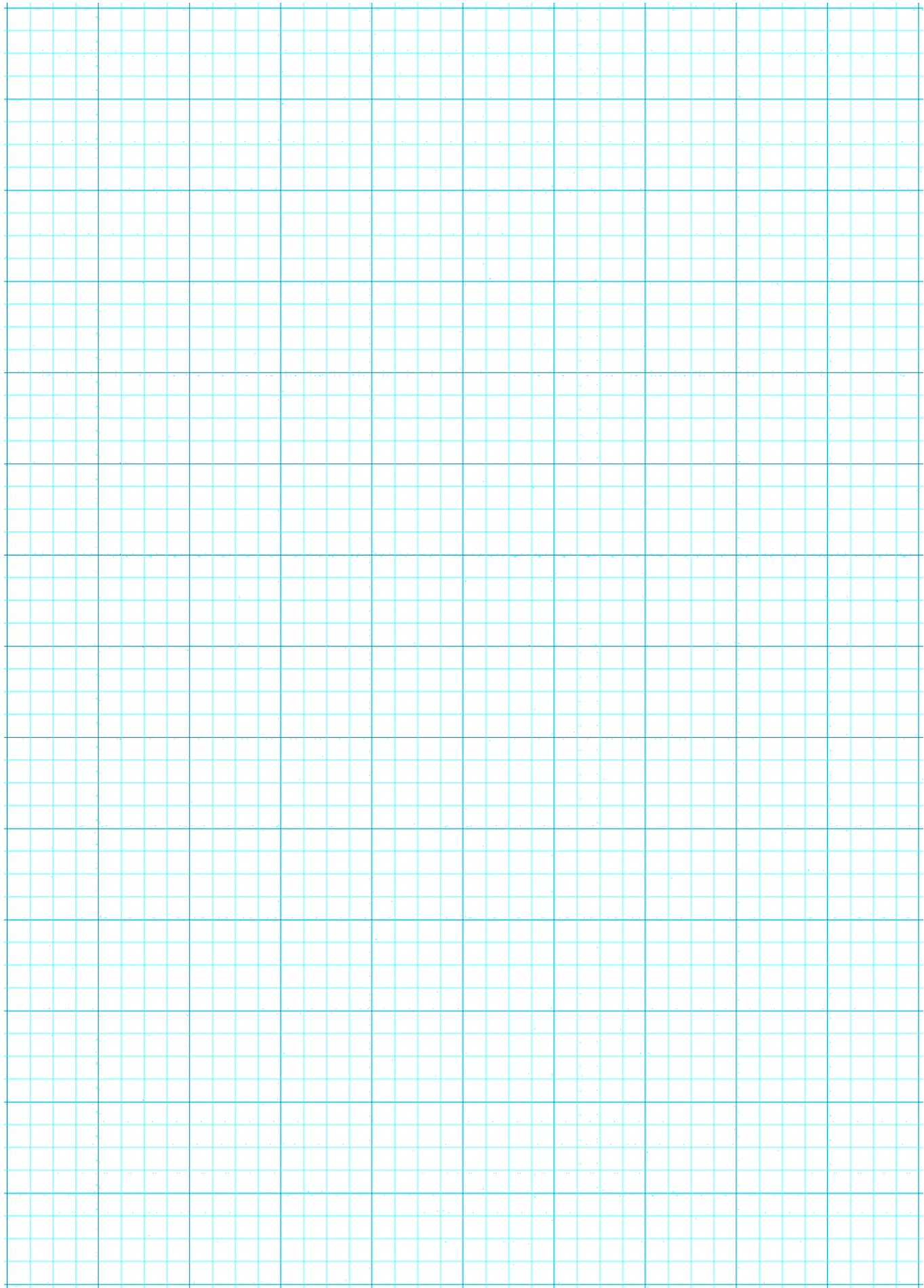
.....

6. Write out the definition of the process

.....

.....





Conclusion

1. Describe the relationship between sugar solution and mass of the potato shown in your graph.

.....

.....

.....

.....

2. Explain the relationship between sugar solution and mass of the potato.

.....

.....

.....

.....

Lesson 6: Active Transport

Define active transport:

.....

.....

.....

Watch the video on active transport and answer the following questions:

<https://www.youtube.com/watch?v=eDeCgTRFCbA>

1. What is required in order for active transport to take place?

.....

2. What is the name of the structure which carries molecules across the cell membrane during active transport?

.....

3. Which process released the energy needed for active transport?

.....

4. Give an example of a plant cell where active transport commonly takes place.

.....

5. Which organelle is prevalent (more common) in cells that carry out active transport more frequently?

.....

Comparing diffusion, osmosis and active transport.

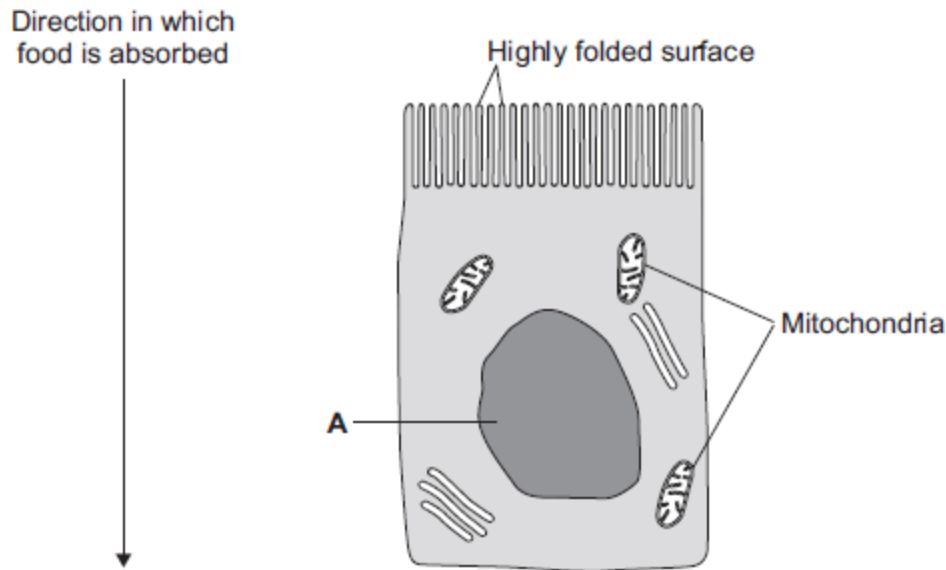
Task: Sort the following statements into the correct columns.

- Passive process
- Active process
- Requires energy
- Does not require energy
- Moves molecules against a concentration gradient
- Move molecules down a concentration gradient
- Moves substances from a high to a low concentration
- Moves substances from a low to a high concentration
- Movement of water only

Diffusion	Osmosis	Active transport

Exam Practice Questions

The image below shows an epithelial cell from the lining of the small intestine.



- (a) (i) In the image above, the part of the cell labelled **A** contains chromosomes.

What is the name of part **A**?

(1)

- (ii) How are most soluble food molecules absorbed into the epithelial cells of the small intestine?

Draw a ring around the correct answer.

diffusion	osmosis	respiration
------------------	----------------	--------------------

(1)

- (b) Suggest how the highly folded cell surface helps the epithelial cell to absorb soluble food.

(1)

- (c) Epithelial cells also carry out active transport.

- (i) Name **one** food molecule absorbed into epithelial cells by active transport.

(1)

- (ii) Why is it necessary to absorb some food molecules by active transport?

(1)

- (ii) Suggest why epithelial cells have many mitochondria.

(2)

- (d) Some plants also carry out active transport.

Give **one** substance that plants absorb by active transport.

(1)

Chicken eggshells contain calcium. Calcium ions are moved from the shell into the cytoplasm of the egg.

Table 2 shows information about the concentration of calcium ions.

Table 2

Location	Concentration of calcium ions in arbitrary units
Egg shell	0.6
Egg cytoplasm	2.1

- (e) Explain how calcium ions are moved from the shell into the cytoplasm of the egg.

(3)

- Q2.** (a) Some scientists investigated the rates of absorption of different sugars by the small intestine.

In one experiment they used a piece of normal intestine.

In a second experiment they used a piece of intestine poisoned by cyanide. Cyanide is poisonous because it prevents respiration.

The results are shown in the table.

Sugar	Relative rates of absorption	
	Normal intestine	Intestine poisoned by cyanide
Glucose	1.00	0.33
Galactose	1.10	0.53
Xylose	0.30	0.31
Arabinose	0.29	0.29

- (i) Name **two** sugars from the table which can be absorbed by active transport.

1. _____

2. _____

(1)

- (ii) Use evidence from the table to explain why you chose these sugars.

(3)

- (b) All of the sugars named in the table can be absorbed by diffusion. Explain how information from the table provides evidence for this.

(2)

Lesson 7: Pathogens

TASK 1: Sort the diseases into the table. Is it a communicable or non-communicable disease?

Tuberculosis (air borne - bacteria)	Diabetes (pancreas malfunction)
Respiratory Diseases (e.g. asthma) (genetics/ environmental)	Cancer (genetics/ environmental)
Arthritis (genetics/ environmental)	Cholera (water borne - bacteria)
Measles (air borne - virus)	Stroke (genetics/ environmental)
Small pox (body fluids - virus)	Chicken pox (air borne & fluids - virus)
High blood pressure(genetics/ environmental)	Influenza (air borne - virus)
Typhoid (water borne - bacteria)	Sickle cell anaemia (genetics)
Heart Disease (genetics/ environmental)	HIV (blood and fluids - virus)

Communicable	Non-communicable

Questions

1. Describe how bacteria make people ill.

.....

2. Give 2 examples of illnesses caused by fungi.

.....

3. Describe how viruses make people ill.

.....

4. Give 2 examples of illnesses caused by protists.

.....

5. Explain why a virus is not a living organism.

.....

Diseases and Their Causes

Disease	Pathogen	How it's spread?	Symptoms	Treatment
Flu				
Pneumonia				
Ringworm				
Cholera				
Tobacco Mosaic Virus				
Malaria				
HIV (AIDS)				

Cells and Microbes – Summary

- Living things are made from cells and carry out the processes of life
- There are 5 kingdoms – animals, plants, protoctists, fungi and bacteria
- Viruses are not living organisms as they do not have a cell surface membrane, cytoplasm or other organelles. They are parasites that must rely on a host to reproduce.
- Animal cells are different from plant cells as they do not have a cellulose cell wall, chloroplasts and a large permanent vacuole
- In a multicellular organism, there are differentiated cells each adapted for a specific function
- Stem cells are undifferentiated but with the right stimulus have the potential to become any specialized cell
- Molecules can move in and out of cells using the processes of diffusion, osmosis and active transport
- Diffusion is passive - molecules from an area of high concentration to an area of low concentration until dynamic equilibrium is reached.
- Osmosis is also passive but applies only to water, moving from an area of high water concentration (dilute solution) to an area of low water concentration (concentrated solution) through a selectively/partially permeable membrane
- Active transport requires energy and moves substances against the concentration gradient e.g. the movement of mineral ions into root hair cells from the soil
- The rate of diffusion is affected by: temperature, the surface area to volume ratio, the distance molecules have to travel and the difference in the concentration gradient
- Most multicellular organisms need to rely on a transport system to deliver substances to all cells as diffusion would not be efficient enough to support life
- Fungi can be unicellular like yeast or multicellular like mushrooms and mould
- All fungi are saprotrophs – they feed using extracellular digestion
- Fungal cells have a cell wall made from chitin, mycelium made of hyphae but cannot photosynthesize.
- Fungi have an important roles as a food source, in food production (bread and beer) and are essential for recycling nutrients (decomposers)
- Microbes that cause infectious diseases are called pathogens and may be a virus, bacterium, fungus or a protoctist.
- Bacteria have a cell wall, some can photosynthesize but they do not have a nucleus. Their DNA is circular and is found in the cytoplasm. They also contain plasmids.

Unit 1 Cells and Microbes Revision Notes

Syllabus Statements	Revision Notes
1a) Characteristics of living organisms 1.1 Understand that living organisms share the following basic characteristics: • they require nutrition • they respire • they excrete their waste • they respond to their surroundings • they move • they control their internal conditions • they reproduce • they grow and develop.	
1b) Variety of living organisms 1.2 describe the common features shown by eukaryotic organisms: plants, animals, fungi and protoctists 1.3 describe the common features shown by prokaryotic organisms such as 1.4 Recall the term 'pathogen' and know that pathogens may be fungi, bacteria, protoctists or viruses.	Plants: Animals: Fungi: Protoctists: Bacteria: Viruses:
2a) Level of organisation 2.1 describe the levels of organisation within organisms: organelles, cells, tissues, organs and systems.	
2b) Cell structure 2.2 describe cell structures, including the nucleus, cytoplasm, cell membrane, cell wall, mitochondria, chloroplasts, ribosomes and vacuole 2.3 describe the functions of the nucleus, cytoplasm, cell membrane, cell wall, mitochondria, chloroplasts, ribosomes and vacuole 2.4 know the similarities and differences in the structure of plant and animal cells.	

2d) Movement of substances into and out of cells 2.15 understand the processes of diffusion, osmosis and active transport by which substances move into and out of cells 2.16 understand how factors affect the rate of movement of substances into and out of cells, including the effects of surface area to volume ratio, distance, temperature and concentration gradient 2.17 Practical: investigate diffusion and osmosis using living and no-living systems 2.51 understand why simple, unicellular organisms can rely on diffusion for movement of substances in and out of the cell	
--	--

Revision - Knowledge Recall Multiple Choice Test

1. Which of the following sets of conditions would make diffusion of a substance into a cell fastest?

	cell surface area	concentration difference
A	high	high
B	high	low
C	low	high
D	low	low

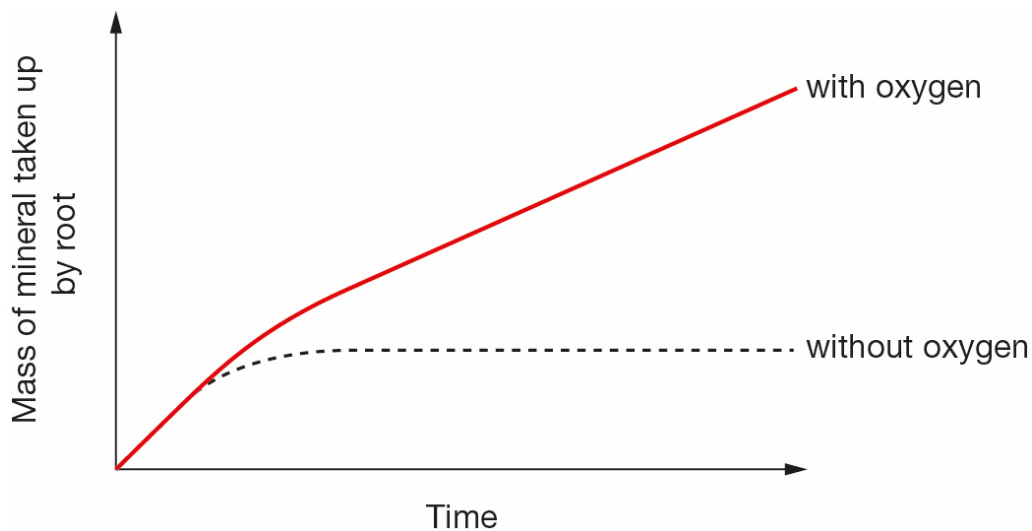
2. Here are some statements about active transport.

1. active transport pumps particles down a concentration gradient
2. active transport requires ATP
3. active transport requires living cells

Which of the statements are correct?

- A 1 and 2
- B 1 and 3
- C 2 and 3
- D 1, 2 and 3

3. The graph shows the transport of mineral ions into a plant root hair cell over time with oxygen and without oxygen.



Which processes does the root hair cell use to transport the mineral ions?

- A active transport alone
- B active transport and diffusion
- C diffusion and osmosis
- D diffusion alone

4. Which of the following is the correct order of structures?

- A cell, tissue, system, organ
- B cell, tissue, organ, system
- C cell, organ, system, tissue
- D cell, system, tissue, organ

5. Which row correctly states the functions of the organelles?

	mitochondria	ribosomes	chloroplasts
A	photosynthesis	protein synthesis	respiration
B	photosynthesis	respiration	protein synthesis
C	respiration	photosynthesis	protein synthesis
D	respiration	protein synthesis	photosynthesis

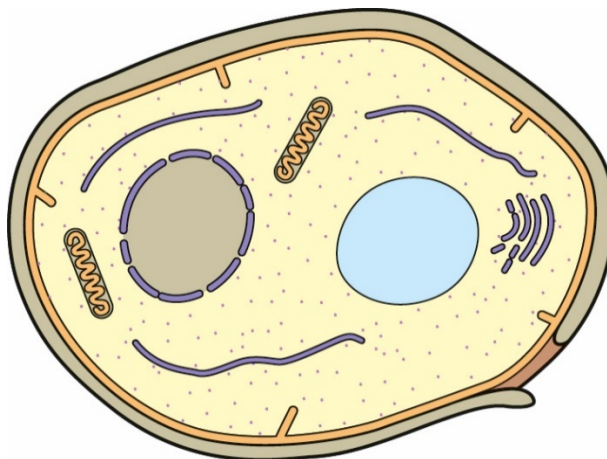
6. Which of these structures are found in both plant and animal cells?

- A cell membrane, cell wall and large vacuole
- B cell membrane only
- C cell wall and large vacuole
- D cell wall only

7. Which row correctly identifies a prokaryotic cell?

	nucleus	mitochondria
A	absent	absent
B	absent	present
C	present	absent
D	present	present

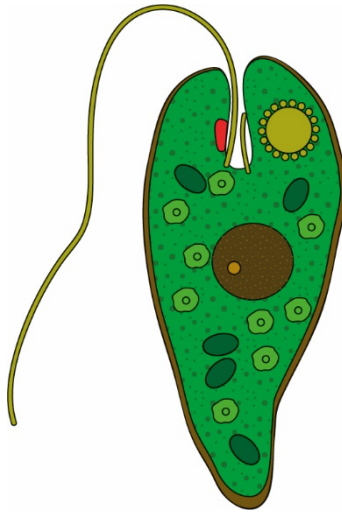
8. The diagram shows a yeast cell.



9. Which of the following statements about yeast is correct?

- A it is a eukaryote and has a cell wall made of cellulose
- B it is a eukaryote and has a cell wall made of chitin
- C it is a prokaryote and has a cell wall made of cellulose
- D it is a prokaryote and has a cell wall made of chitin

10. The diagram shows a single celled organism called *Euglena*.



In which kingdom should *Euglena* be placed?

- A animals
- B fungi
- C plants
- D protocists

11. Which of the following can have both multicellular and single-celled organisms?

- A animals
- B bacteria
- C fungi
- D plants

12. Which of the following rows matches the carbohydrate molecule with a correct group?

	glycogen	starch
A	animals	fungi
B	fungi	plants
C	plants	fungi
D	plants	animals

13. Below is a list of the characteristics of life.

movement
respiration
sensitivity
growth
reproduction
excretion
nutrition
control

How many of these characteristics of life do bacteria exhibit?

- A 5
- B 6
- C 7
- D 8

14. Which of the following statements about viruses is correct?

- A viruses are made of cells
- B viruses can reproduce outside the body
- C viruses carry DNA or RNA
- D viruses release energy by respiration

15. Which of the following groups contain examples of pathogens?

	bacteria	protocists	viruses
A	no	no	yes
B	no	yes	no
C	yes	yes	yes
D	yes	no	yes

16. Below are some statements about fungi.

- fungal hyphae release enzymes to digest food
- the network of hyphae is called a mycelium
- many fungi use saprophytic nutrition
- examples of fungi include yeast and *Mucor*

How many of the statements are correct?

- A 1
- B 2
- C 3
- D 4

Revision Exam: Application of Knowledge

1 a) Although plants and animals have many different features, they also have some features in common.

The table lists some features.

In each box, place a tick(✓) if the feature is present or a cross (x) if the feature is absent. One has been done for you. (4)

Feature	Plants	Animals
Can move from place to place	x	✓
Can carry out photosynthesis		
Are multicellular		
Have cells with cell walls		
Store carbohydrate as glycogen		

(b) Organisms that cause disease are known as pathogens.

Give two groups of organism that include pathogens.

(2)

.....

.....

(Total for Question 1= 6 marks)

2. The table lists the names of different structures found within a human.

Complete the table by numbering each structure in order of its size.

Use number 1 for the smallest structure through to number 4 for the largest.

Name of structure	Order of size
Brain	
Nucleus	
Nervous system	
Nerve cell	

(Total 3 marks)

3. John wanted to investigate the effect of the size of potato tissue on the rate of osmosis.

He cut three different sized cubes of potato, one $0.5 \times 0.5 \times 0.5$ cm, one $1 \times 1 \times 1$ cm and one $2 \times 2 \times 2$ cm.

He weighed the potato cubes and recorded their masses.

He then placed each cube into a beaker of distilled water and left them for 1 hour.

He weighed them again and recorded their new masses.

In each case the mass of the potato cubes increased.

The table shows his results.

Potato cube size in cm	Original mass in g	Final mass in g	Percentage change in mass (%)
$0.5 \times 0.5 \times 0.5$	0.06	0.07	16.67
$1 \times 1 \times 1$	0.51	0.56	
$2 \times 2 \times 2$	4.04	4.10	1.49

His teacher told him to calculate the percentage change in mass of each cube.

(a) (i) Calculate the percentage change in mass for the $1 \times 1 \times 1$ cm cube.
Show your working.

(2)

Answer:.....

(ii) Explain why John converted change in mass to percentage change in mass. (1)

.....

(b) Use your knowledge of osmosis to explain why the mass of each cube increased. (3)

.....

(c) John’s teacher suggested that the different sized cubes have a different surface area and volume.

Use the formulae below to calculate the surface area (SA) of each cube, the volume (Vol) of each cube and their SA:Vol ratios.

SA= 6 x (cube side in cm)²

Vol= (cube side in cm)³

Write your answers in the table below. One cube has been done for you. (3)

Cube side in cm	Surface area in cm ²	Volume in cm ³	Sa:Vol ratio
0.5	1.5	0.125	12
1.0			
2.0			

(d) Explain the effect of the different SA:Vol rations on the rate of osmosis into the potato. (2)

.....

.....

.....

.....

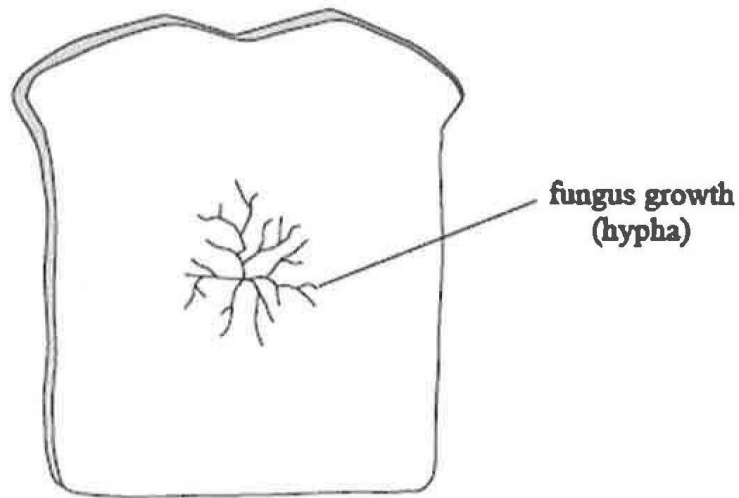
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(e) Potato tissue is made from plant cells.
Draw and label a plant cell in the space below.

(3)

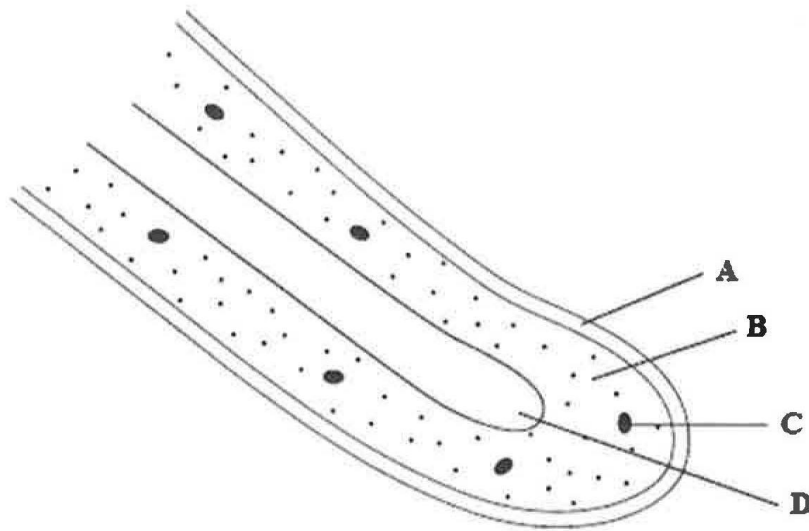
(Total for Question 3 =11 marks)

4. The drawing shows a piece of bread. The bread is going mouldy because a fungus is growing and feeding on it. When the fungus grows it produces many threads (hyphae) that spread over the bread.



The hyphae secrete enzymes that digest the bread. The products from this digestion are then absorbed by the fungus.

- (a) The diagram shows a magnified view of a single hypha from the fungus.

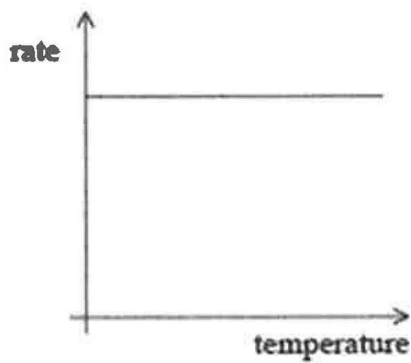


- (i) Name the parts labelled A, B, C and D.

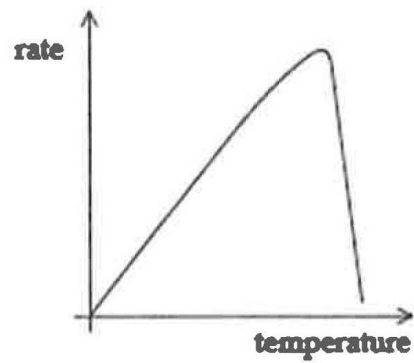
A:
B:
C:
D:

(Total for Question 4 = 4 marks)

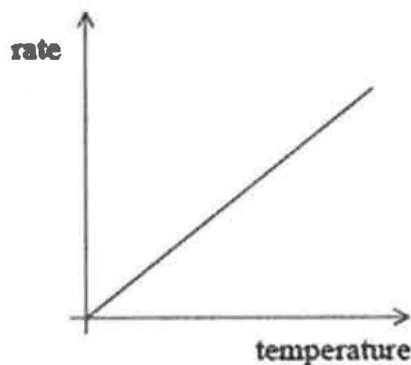
5. The graphs show the effect of changing the temperature on the rate of movement of substances into cells.



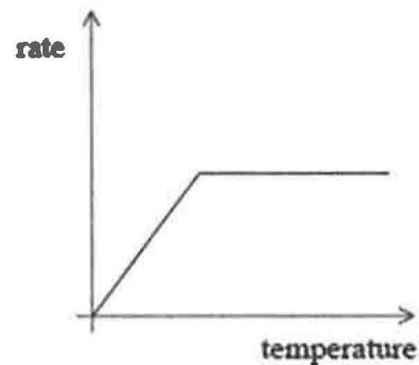
A



B



C



D

- (i) Which graph shows the effect of temperature on diffusion? (1)

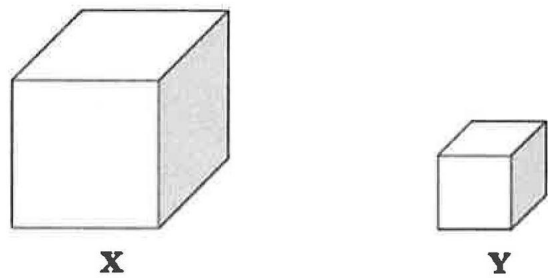
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- (ii) Which graph shows the effect of temperature on active transport? (1)

.....

(Total for Question 5= 2 marks)

6. The diagram shows two different-sized potato cubes, X and Y, that were placed in distilled water for one hour.



(a) Explain why both cubes gained in mass after one hour. (3)

.....

.....

.....

.....

(b) (i) Put a tick ✓ in the row of the table that correctly describes the change in mass for cube X compared to cube Y.

Mass of water absorbed in g	% increase in mass	tick
more	lower	
less	lower	
less	higher	
more	higher	

(ii) Calculate the percentage increase in length of the potato cylinders after one hour in water. Show your working. (2)

.....

Answer:%

(iii) Explain why the potato cylinders increased in length when they were in water. (3)

.....

.....

.....

.....

c) Suggest why the graph levelled out for solutions of sucrose greater than 5 arbitrary units. (1)

.....

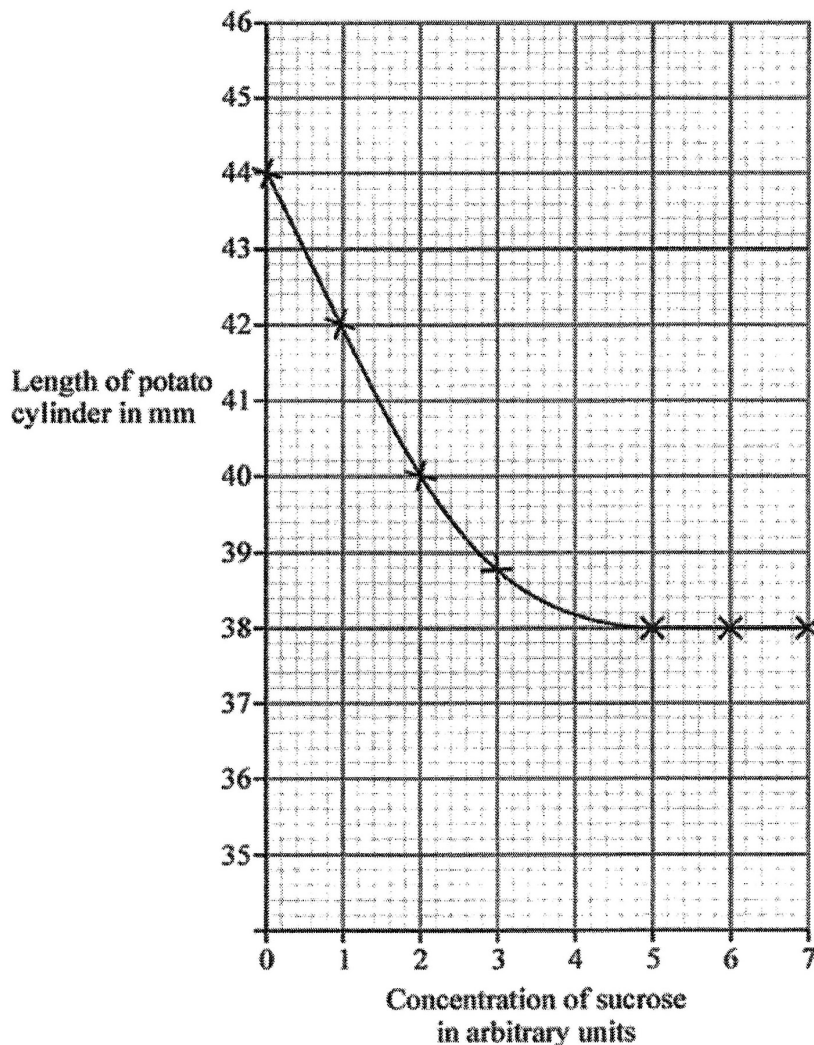
7. A group of students investigated osmosis.

They cut potato cylinders, 40 mm in length.

Some of these cylinders were put into water and some into sucrose solutions of different concentration.

After one hour the length of each potato cylinder was measured.

The graph shows the length of the cylinders after one hour plotted against the concentration of sucrose.



Use the graph and your knowledge to answer the questions.

- (a) (i) which concentration of sucrose resulted in exactly 2 mm increase in length of potato cylinders? (1)

- (ii) Which solution of sucrose had the same concentration as the cell sap in the potato cells? Give a reason for your answer. (2)